

INTRODUCTION TO DERMATOLOGY

Atif Saleem, DO

With contributions from Liisa A. Russell, MD,
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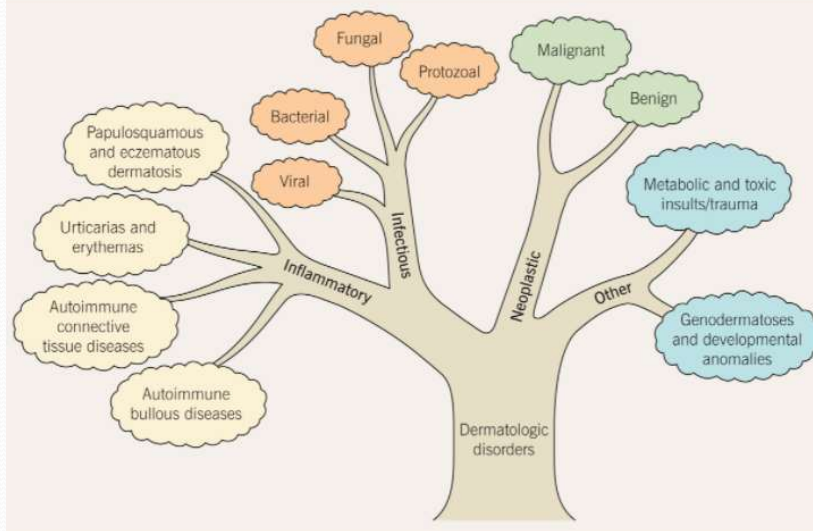
OBJECTIVES

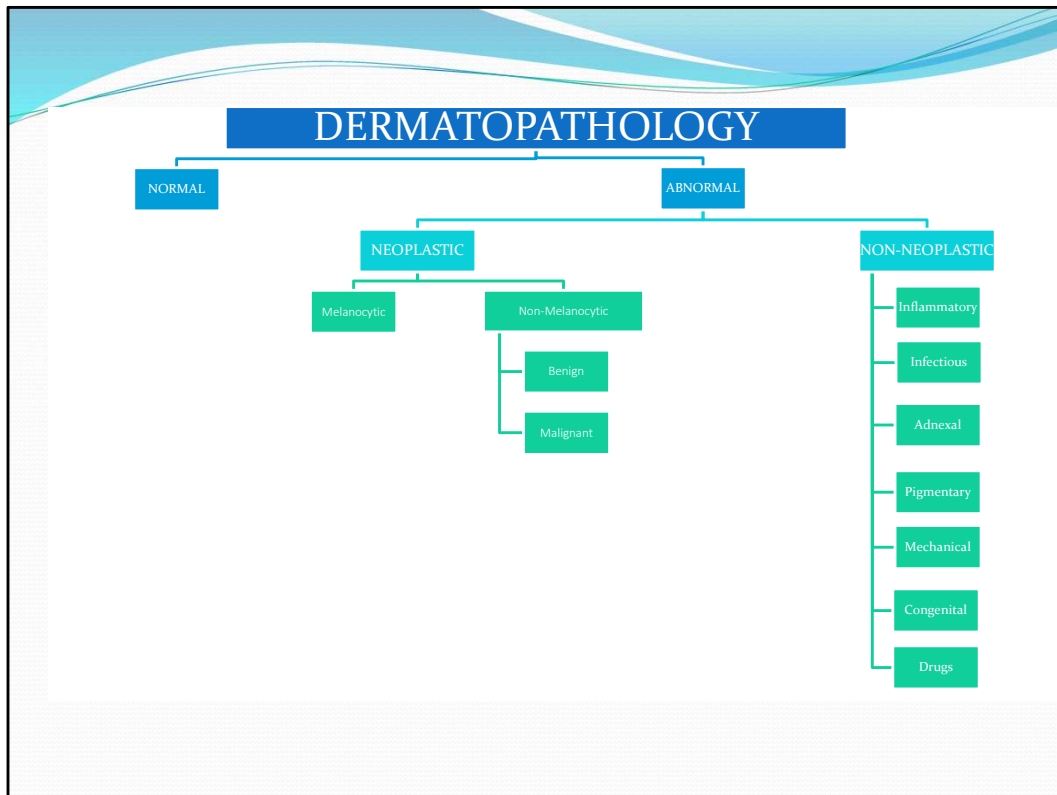
- Recognize the principles of a comprehensive skin examination
- Recognize the components of normal skin
- Recognize the difference between primary and secondary lesions.
- Identify the following types of skin lesions from descriptions and/or photographs
 - Excoriation, erosion, ulcer, fissure, lichenification, macule, patch, papule, nodule, plaque, vesicle, bulla, blister, pustule, cyst, wheal, scale, crust, atrophy, scar
- Identify the correct usage of the descriptive language:
 - Discoid, nummular, arcuate, annular, livedo, reticulate, target, linear, serpiginous

RECOMMENDED READING FOR THIS MODULE

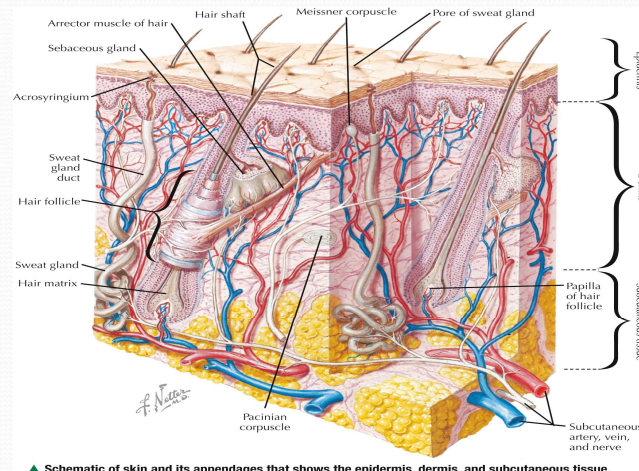
- Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology, 9th Edition (2023)
- Robbins & Kumar Basic Pathology, 11th edition, chapter 22 (2023)

CLASSIFICATION SCHEME FOR DERMATOLOGIC DISORDERS





NORMAL



▲ Schematic of skin and its appendages that shows the epidermis, dermis, and subcutaneous tissue.

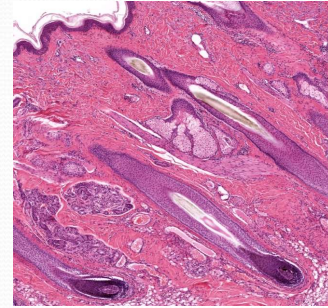
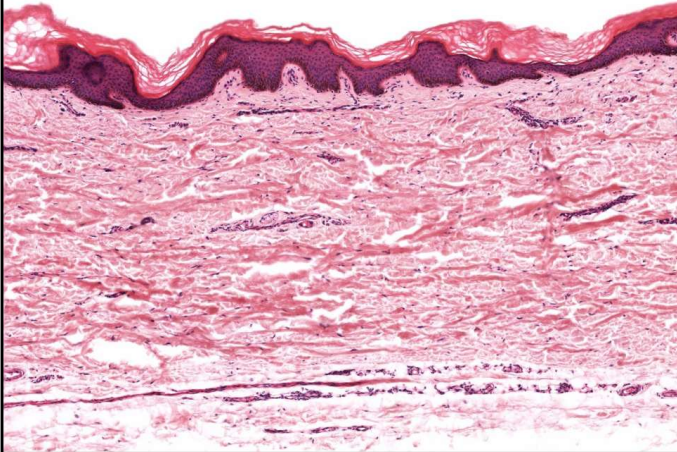
Integumentary System

Ovalle, William K., PhD, Netter's Essential Histology, 11, 263-283

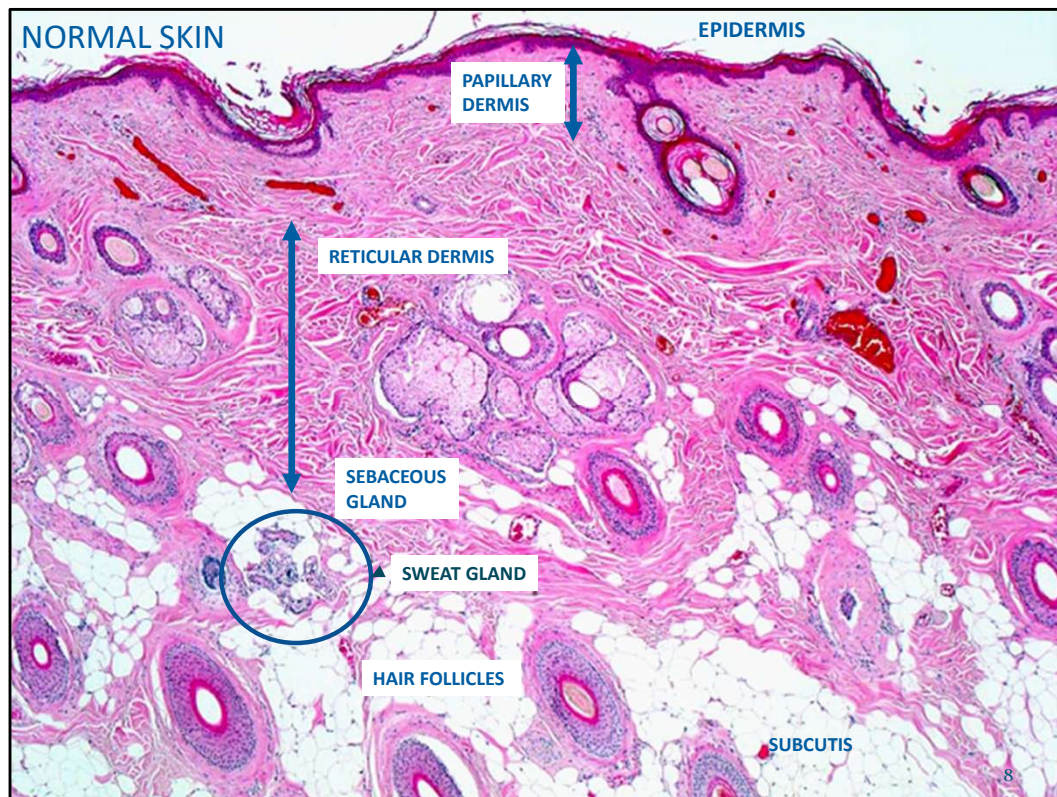
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NORMAL

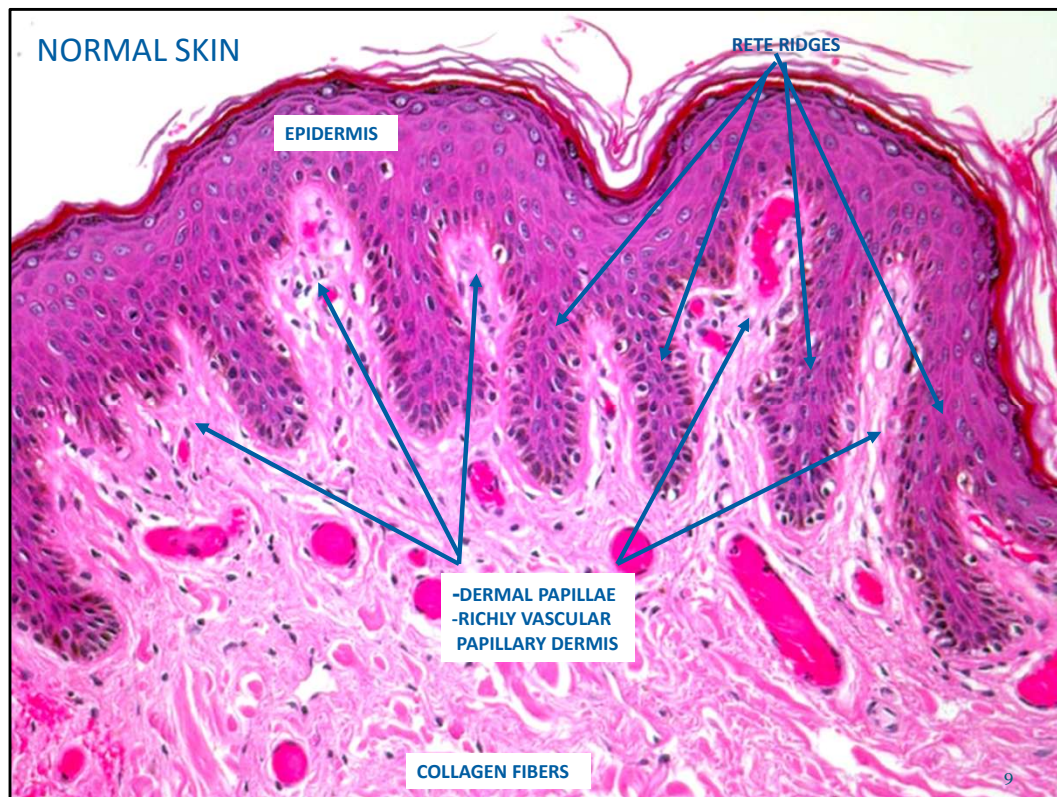


We will annotate this slide during the lecture

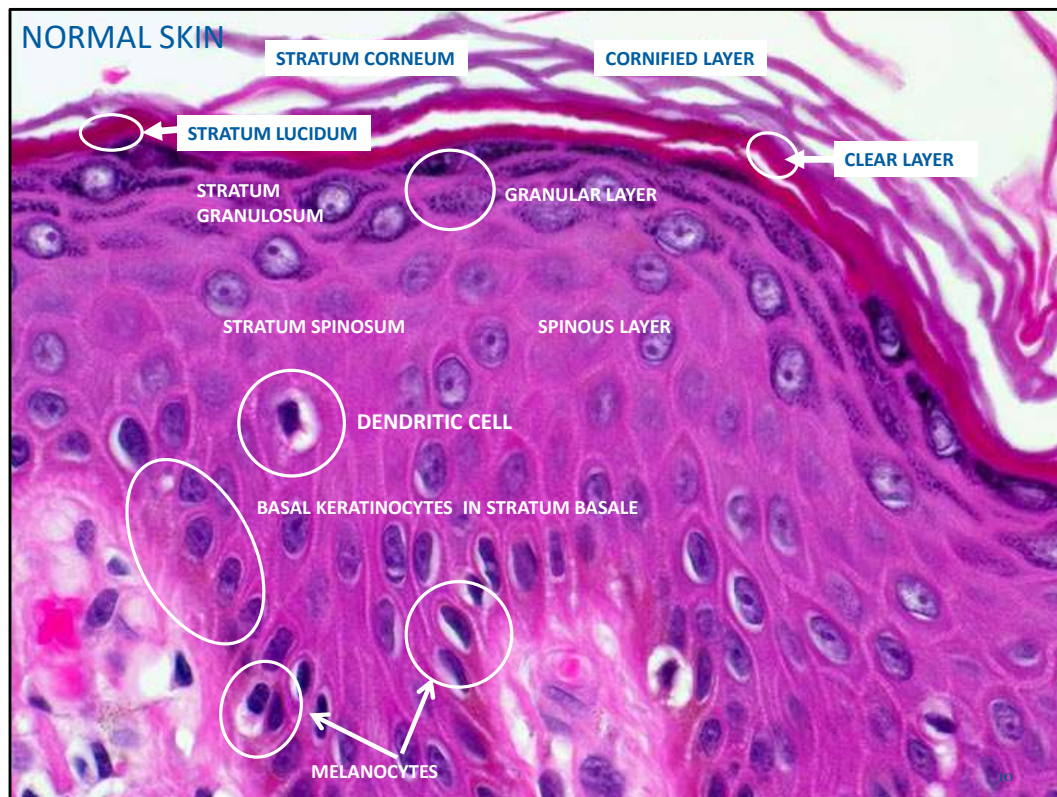


Normal skin.

From Dr. Liisa Russell.



Epidermis and papillary dermis



Normal skin

From Dr. Liisa Russell.

THE CHALLENGE OF INTERPRETING SKIN FINDINGS ON EXAMINATION

- What is the challenge?
 - There are many skin diseases
 - A lesion can vary in appearance from one person to the next
 - Skin diseases are dynamic
- It may be necessary to learn to recognize hundreds of permutations in order to be a confident diagnostician
- It is important to have a **methodical approach** when examining skin lesions

HISTORY OF SKIN LESIONS

SEVEN KEY QUESTIONS

- **When?** Onset/duration
- **Where?** Site of onset
- **Does it itch or hurt?** Symptoms
- **Has it spread (pattern of spread)?** Extent
- **Have individual lesions changed? How?** Evolution
- **Provocative factors?** Heat, cold, sun, exercise, foods, medications/drugs, hormonal status (e.g. pregnancy), season, travel, etc.
- **Previous treatment(s)?** Topical and systemic

FOCUSED HISTORY



- **Constitutional symptoms**
 - "Acute illness" syndrome: headaches, chills, feverishness, weakness
 - "Chronic illness" syndrome: fatigue, weakness, anorexia, weight loss, malaise
- **General history of present illness as indicated by clinical situation**
 - Attention must be paid to constitutional and prodromal symptoms
- **Past medical history**
 - Operations, prior skin lesions/rashes, illnesses (hospitalizations?), allergies, medications (current and past), habits (smoking, alcohol intake, drugs), history of atopy (asthma, hay fever, eczema)

FOCUSED HISTORY



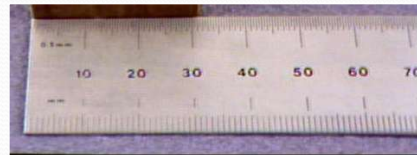
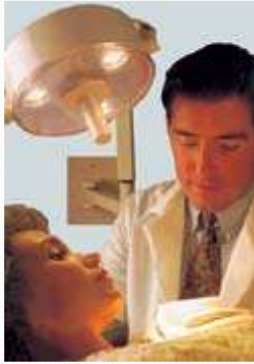
- **Family medical history:**
 - Psoriasis, atopic dermatitis, melanoma, xanthomas, tuberous sclerosis
- **Social history:**
 - Occupation, hobbies, exposures, travel, recreational drug use
- **Sexual history:**
 - HIV risk factors: IV drugs, sexually active, multiple partners, history of sexually transmitted diseases?
- **ROS:**

EXAMINATION OF THE SKIN

- Important to examine **more than enough**, rather than to examine **less than enough** skin
- Examination of the **entire body** is necessary, if the patient has a **diffuse skin eruption**
 - Examine the entire skin surface
 - Include mucous membranes, genital and anal regions, hair and nails as well as lymph node stations in the examination
 - Pay attention to distribution of individual lesions
 - Use hand lens and adequate illumination
 - ***Palpate***

EXAMINATION OF THE SKIN

- Always have the right tools handy to do the job!



EXAMINATION OF THE SKIN

- **Proper Lighting**
- **Oblique Lighting**
 - Subtle elevation/depression can be detected
- **Wood's lamp** (ultraviolet long-wave light, "black" light)
 - Fluorescent pigments and subtle color differences of pigmentation can be visualized
- **Magnification**
- **Diascopy (dermoscopy)** (firmly pressing a microscopic slide or a glass spatula over a skin lesion)
 - Helps determine whether the red color of a macule or papule is due to **blanching erythema** (capillary dilatation) or **non-blanching purpura** (extravasation of blood)



VASCULAR PATTERN BY DERMOSCOPY

INTRADERMAL NEVUS



Comma-shaped vessels

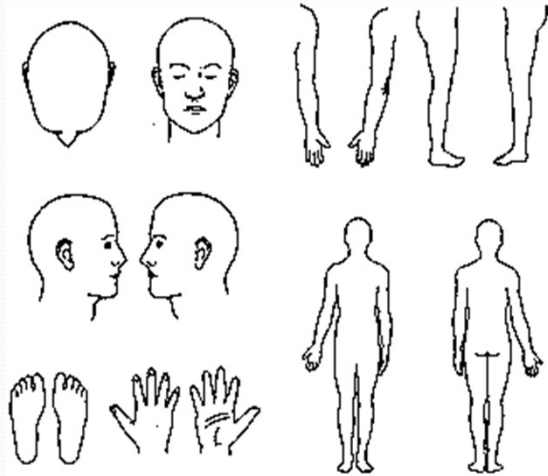
MALIGNANT MELANOMA, 3.0 MM



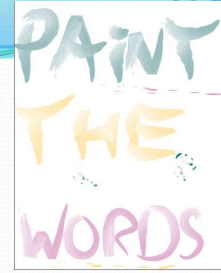
Irregular vessels
Pinpoint vessels

EXAMINATION OF THE SKIN

- Learn to draw:
 - Sketch the eruption, especially its distribution
 - Use preprinted figures for written notes



THE LANGUAGE OF DERM: PAINT A PICTURE WITH WORDS



- This is a hard-earned, invaluable skill that takes a real effort to master
- Someday soon you will be challenged with trying to describe a strange rash over the phone to a dermatologist
- That is when you will be thankful to have learned the descriptive language that is at the root of successful communication with your colleagues

LEARN THE LANGUAGE OF DERMATOLOGY

- It is essential that you describe the skin lesions accurately
- This allows physicians seeing the patient in follow-up to get a clear picture of what you observed
 - It will be possible to determine if the lesions have improved, worsened, or evolved
- The prerequisite of dermatologic diagnosis is the recognition of
 1. Type of skin lesion
 2. Color
 3. Margination
 4. Consistency
 5. Shape
 6. Arrangement
 7. Distribution

TYPE OF SKIN LESION: PRIMARY

- Physical changes in the skin considered to be caused directly by the disease process
- Rarely specific to a single disease entity

Macule

Papule

Plaque

Nodule

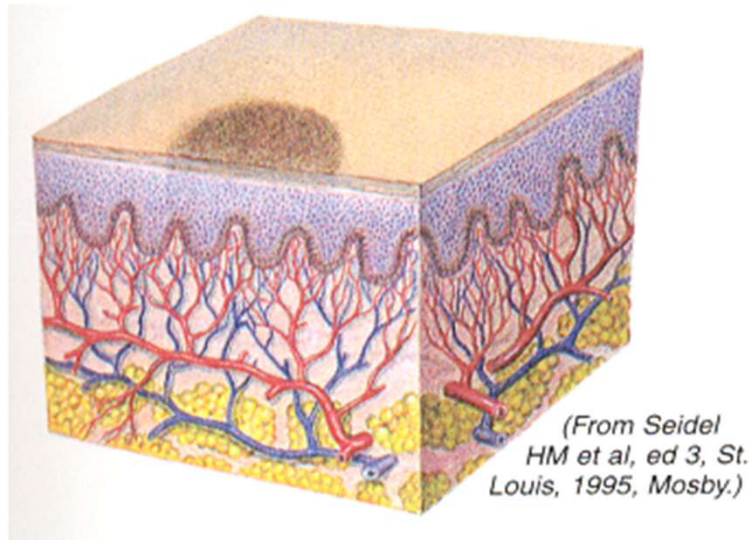
Pustule

Vesicle

Bulla

Wheal

Telangiectasia



MACULE

- A circumscribed, flat area of discoloration <1 cm in diameter
- Color may be brown, blue, red, or white (hypopigmented)

MACULES



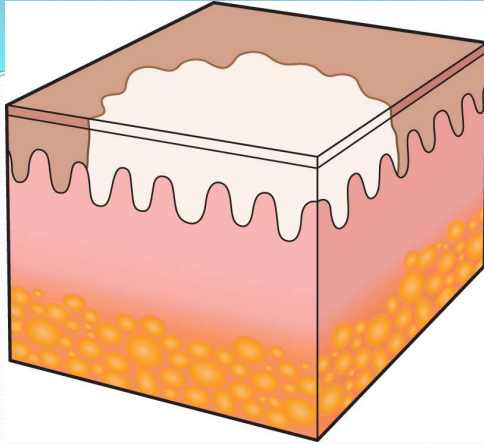
Tuberous sclerosis



Lentigo

- Solar lentigo?
- R/O malignant melanoma in situ

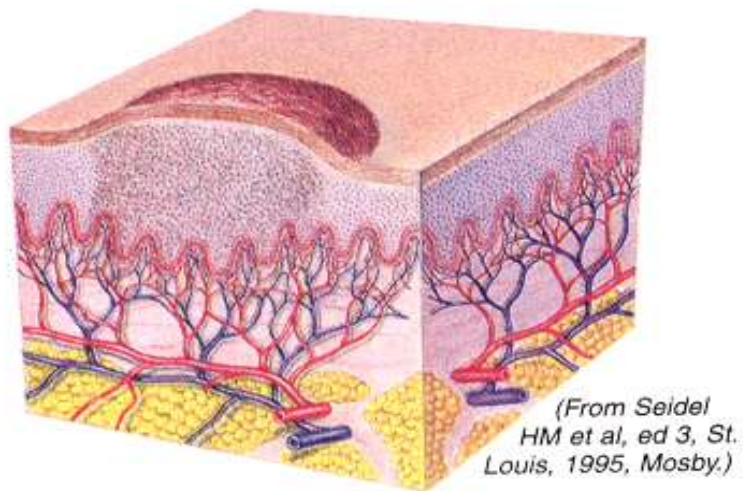
24



Vitiligo

PATCH

- A circumscribed, flat area of discoloration >1 cm in diameter
- Color may be brown, blue, red, or white (hypopigmented)



PAPULE

- An elevated solid lesion, <1 cm in diameter
- Color varies
- Papules may coalesce and form plaques

PAPULES



Molluscum contagiosum



Seborrheic keratosis

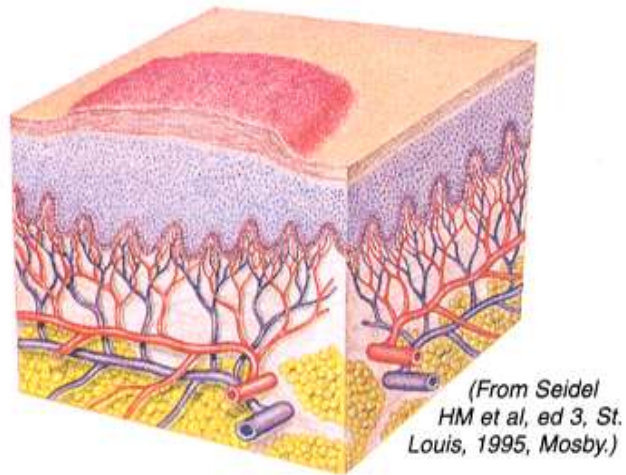
PAPULES



Sebaceous hyperplasia



Verruca vulgaris (wart) with cylindrical projections on the surface due to hyperplasia (acanthosis) of the epidermis



PLAQUE

- An elevated, circumscribed, solid, superficial lesion
- >1 cm in diameter

PLAQUES



Eczema



Psoriasis

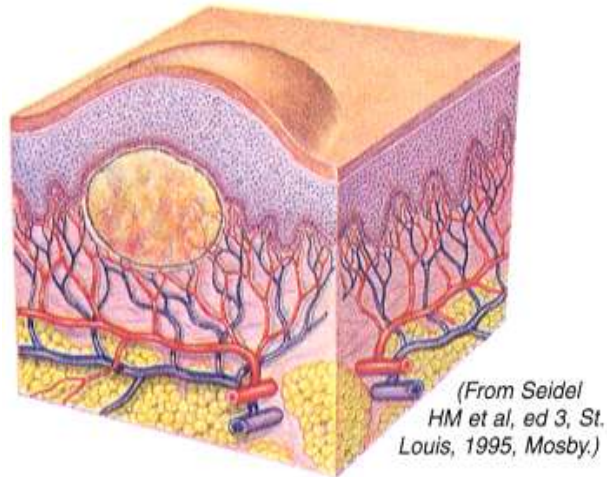
PLAQUES



Lichen planus



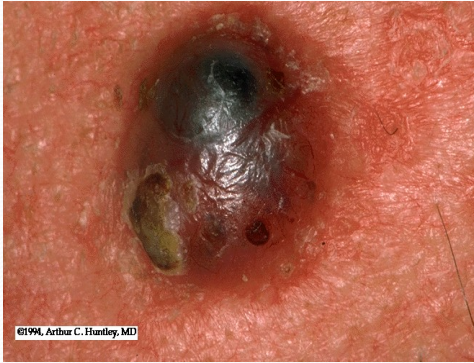
Tinea corporis



NODULE

- A circumscribed, elevated, solid lesion
- > 1 cm in diameter
- A large nodule (>5 cm) is referred as a tumor

NODULES



Nodular pigmented lesion

- Nodular melanoma



Skin-colored nodules

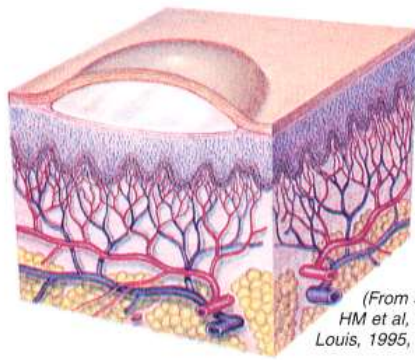
- May arise from the dermis or subcutaneous connective tissue

TUMOR



34

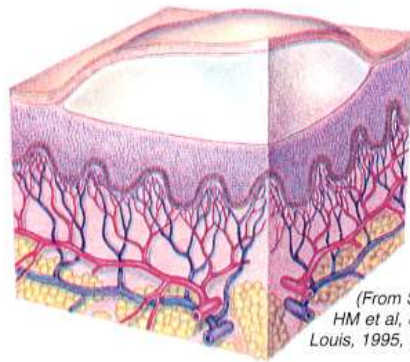
In this case: well demarcated lipoma.



(From Seidel
HM et al, ed 3, St.
Louis, 1995, Mosby.)

VESICLE

- A circumscribed collection of free fluid
- <1 cm in diameter



(From Seidel
HM et al, ed 3, St.
Louis, 1995, Mosby.)

BULLA

- A circumscribed collection of free fluid
- >1 cm in diameter

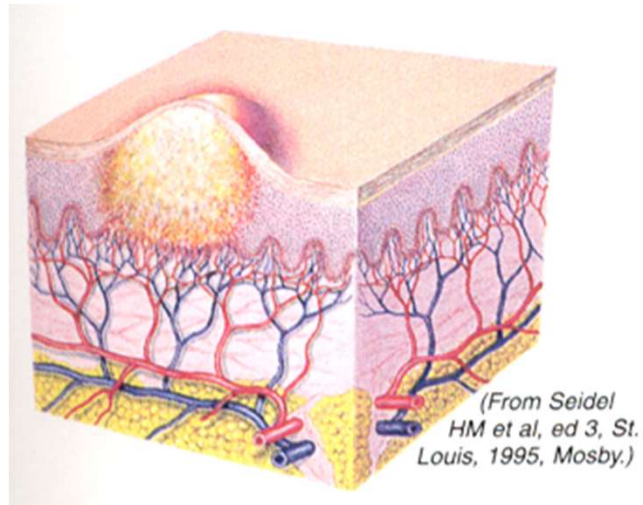
VESICLES



Chicken pox



Bullous pemphigoid



PUSTULE

- A circumscribed collection of free fluid and leukocytes (purulent fluid)
- Usually < 1 cm in diameter

PUSTULES



Folliculitis

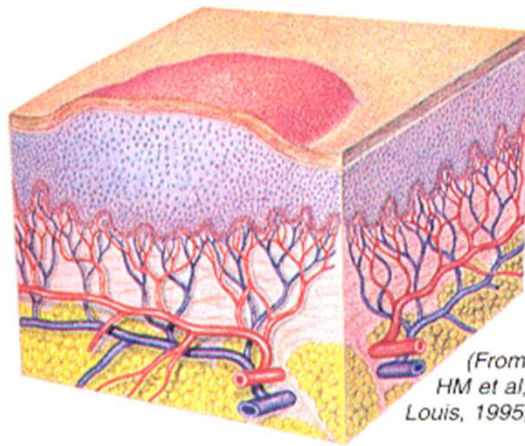


Herpes simplex

VESICLE vs. PUSTULE



Wait for a few hours-a couple of days and the vesicle turns into a pustule.



(From Seidel
HM et al, ed 3, St.
Louis, 1995, Mosby.)

WHEEL (HIVE)

- A firm edematous plaque
- Results from infiltration of dermis with fluid
- Transient; may last only a few hours

WHEELS



Hives



Dermatographism

TELANGIECTASIA



- Permanent dilatation of superficial blood vessels in the skin
- May occur as an isolated phenomenon or as part of a generalized disorder

TYPE OF SKIN LESION: **SECONDARY**

- May evolve from primary lesions, or may be caused by external forces such as scratching, trauma, infection, or the healing process
- May be the only type of lesion present, in which case the primary disease process must be deduced

Scales

Crusts

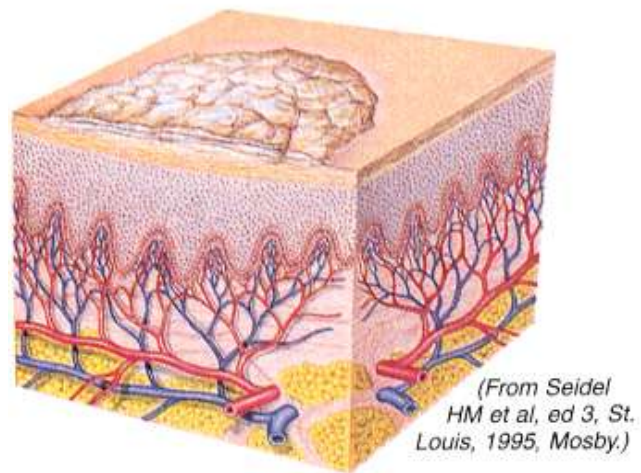
Erosions

Ulcers

Fissures

Atrophy

Scars



SCALES

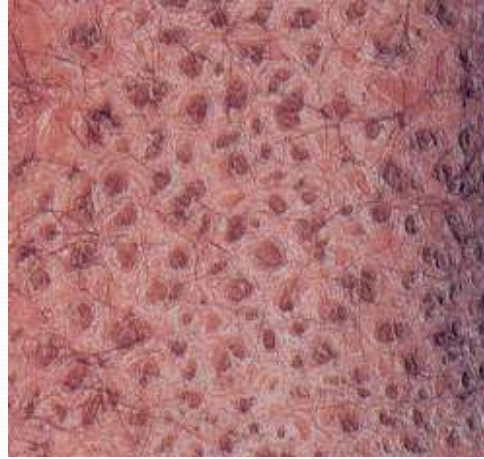
- Excess dead epidermal cells
- Produced by abnormal keratinization and shedding

SCALES



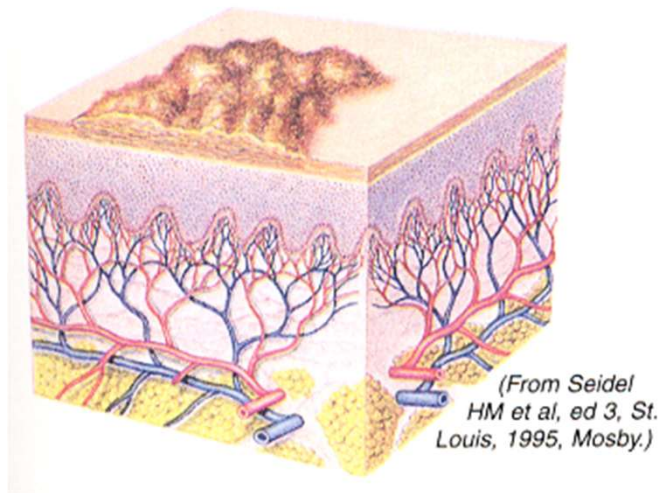
Pityriasis rosea

- Collarette of scales



Ichthyosis vulgaris

- Small quadrangular scales
- Inherited (autosomal semidominant)
- Mutation in the gene encoding filaggrin



CRUST (SCAB)

- A collection of dried serum and cellular debris

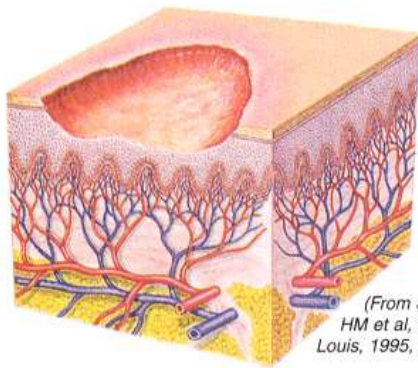
CRUST



Pemphigus foliaceus



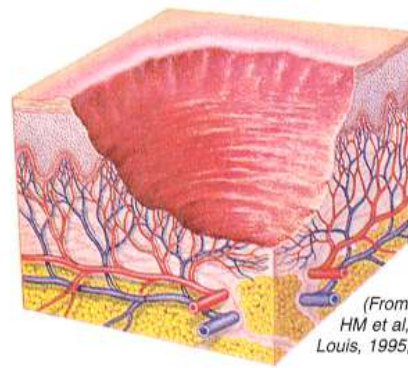
Tinea capitis



(From Seidel
HM et al, ed 3, St.
Louis, 1995, Mosby.)

EROSION

- A focal loss of epidermis
- Does not penetrate below the epidermal-dermal junction
- Heals without a scar



(From Seidel
HM et al, ed 3, St.
Louis, 1995, Mosby.)

ULCER

- A focal loss of both epidermis and dermis
- Heals with a scar

EROSION

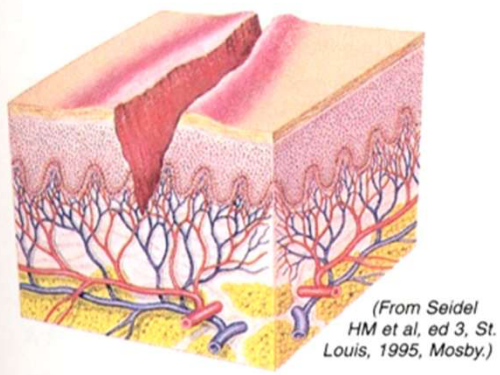


Candidiasis

ULCER



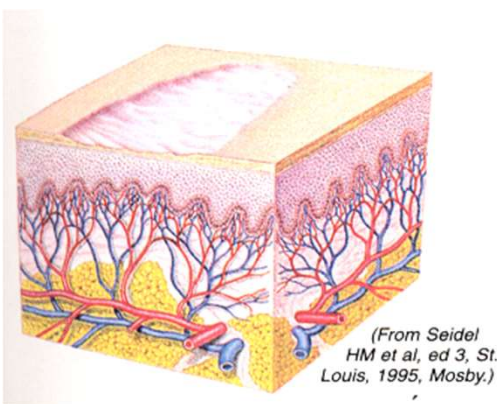
Chancroid



FISSURE

- A linear loss of epidermis and dermis
- Sharply defined, nearly vertical walls

Intertrigo



ATROPHY

- A depression on the skin
- Results from thinning of the epidermis or dermis or both

Striae from excessive topical corticosteroid use

CYST



- A circumscribed lesion with a wall and a lumen
- The lumen may contain fluid or solid matter

DERM TERMS



- **Color**
 - Pink, red, purple, white, tan, brown, black, blue, grey, yellow, hyper-pigmented, hypo-pigmented
 - The color can be uniform or variegated
- **Margination**
 - Well defined (can be traced with the tip of a pencil), ill-defined
- **Shape**
 - Round, oval, polygonal, polycyclic, annular (ring-shaped), iris, serpiginous (snakelike), umbilicated

FITZPATRICK SCALE OF SKIN PHOTOTYPES		
Skin phototype	Skin color	Response to UV irradiation
I	White	Always burns, does not tan
II	White	Burns easily, tans with difficulty
III	Beige	Mild burns, tans gradually
IV	Brown	Rarely burns, tans easily
V	Dark brown	Very rarely burns, tans very easily
VI	Black	Never burns, tans very easily

Table 0.6 Fitzpatrick scale of skin phototypes.

COLOR AS A CLUE TO THE CLINICAL DIAGNOSIS			
Color	Examples of diseases with this color	Color	Examples of diseases with this color
Pink/red Erythema (pink to red-brown, depending upon the skin phototype) 	• Dermatitis • Psoriasis • Morbilliform drug eruption • Viral exanthem • Any condition that causes vasodilation	Purple (violaceous) 	• Purpura, non-palpable (e.g. solar purpura) • Purpura, palpable (e.g. small vessel vasculitis) • Vascular neoplasm (e.g. angiokeratoma, angiosarcoma) • Lichen planus • Lymphoma cutis • Pyoderma gangrenosum – border • Morphea – border (lilac)
Morbilliform (exanthematous) drug eruption 		Palpable purpura of cutaneous small vessel vasculitis 	
Black 	• Necrosis of the skin due to: • Vasculitis (e.g. granulomatosis with polyangiitis) • Thrombosis (e.g. DIC, monoclonal cryoglobulinemia) • Emboli (e.g. ecthyma gangrenosum) • Vasospasm (e.g. severe Raynaud phenomenon) • Vascular compromise (e.g. atherosclerosis, calciphylaxis) • Eschar (e.g. anthrax) • Cutaneous melanoma • Traumatic tattoo (e.g. asphalt)	White 	• Absence of melanocytes or melanin production (e.g. vitiligo, piebaldism, OCA1A) • Scarring (e.g. scarring in discoid lupus erythematosus) • Vasospasm (e.g. Raynaud phenomenon, nevus anemicus) • Deposits (e.g. calcinosis cutis, gouty tophi) • Macerated stratum corneum – mucosal surfaces (e.g. leukoplakia)
Necrosis secondary to vasculopathy from levamisole-contaminated cocaine 		Calcinosis cutis (systemic sclerosis) 	
Blue (periorbital) 	• Dermal melanocytosis (e.g. Mongolian spot, nevus of Ota) • Dermal melanocytoma (e.g. blue nevus) • Cyanosis • Echinosis • Venous congestion (e.g. venous malformation) • Drugs/deposits (e.g. minocycline, traumatic tattoo)	Green 	• Pseudomonas infection • Tattoo • Chloroma • Green hair due to copper deposits
Dermal melanocytosis 		Onycholysis with secondary Pseudomonas infection 	
Brown 	• Pigmented lesions • Lentigo • Seborrheic keratosis • Junctional, compound, or congenital melanocytic nevus • Café-au-lait macule • Dermatofibroma • Melanoma • Pigmented AK, Bowen disease • Postinflammatory hyperpigmentation – epidermal (see Ch. 57) • Melasma • Phytophotodermatitis • Drug-induced hyperpigmentation (e.g. cyclophosphamide) • Metabolic (e.g. Addison disease, hemochromatosis)	Orange-red (salmon) 	• Pityriasis rubra pilaris • Mycosis fungoides (sometimes)
Melasma 		Pityriasis rubra pilaris with islands of sparing 	
Gray 	• Postinflammatory hyperpigmentation – dermal (e.g. erythema dyschromicum perstans; see Ch. 67) • Drugs/deposits (e.g. argyria, chrysalis) • Combined melanocytic nevus • Traumatic tattoo • See Blue (above)	Yellow 	• Solar elastosis • Carotenoderma • Xanthomas (e.g. eruptive, xanthelasma) • Xanthogranulomas • Axillary tumors and hyperplasias with sebaceous differentiation • Necrobiosis lipoidica • Capillitis (yellow-brown background) • Deposits/drugs (e.g. tophi, quinine)
Argylia 		Xanthelasma 	

Table 0.5 Color as a clue to the clinical diagnosis. AKs, actinic keratoses; DIC, disseminated intravascular coagulation; OCA1A, oculocutaneous albinism type 1A. Some photographs courtesy Jean L. Bolognia, MD; Ronald Rapin, MD; Julie V. Schaffer, MD; Kathleen Wozniak, MD.

PALPATION OF CUTANEOUS LESIONS

- Soft (e.g. intradermal nevus) versus firm (e.g. dermatofibroma) versus hard (e.g. calcinosis cutis, osteoma cutis)
- Compressible (e.g. venous lake) versus non-compressible (e.g. fibrous papule)
- Tender (e.g. inflamed epidermoid inclusion cyst, angioliipoma, leiomyoma) versus non-tender (e.g. intradermal melanocytic nevus)
- Blanchable (e.g. erythema due to vasodilation) versus non-blanchable (e.g. purpura)
- Rough (e.g. actinic keratosis) versus smooth (e.g. fibrous papule)
- Mobile (e.g. epidermoid cyst) versus fixed to underlying structures (e.g. soft tissue sarcoma)
- Dermal (e.g. dermatofibroma) versus subcutaneous (e.g. erythema nodosum)
- Temperature – normal versus warmer (e.g. cellulitis) versus cooler (e.g. Raynaud phenomenon)
- Other, e.g. thrill, pulsatile

Table 0.4 Palpation of cutaneous lesions.

ADDITIONAL DESCRIPTORS

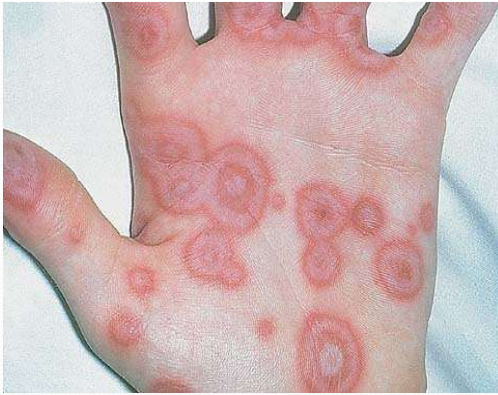
- **Number**
 - Single or multiple lesions
 - Quantify
- **Arrangement**
 - Multiple lesions may be **grouped**:
 - herpetiform, arciform, annular, linear, reticulated (net-shaped), serpiginous (snakelike)
 - **Disseminated**: scattered discrete lesions
- **Confluent**
 - Yes or no
- **Distribution**
 - **Extent**: isolated (single lesions), localized, regional, generalized, universal
 - **Pattern**: symmetric, random, on exposed areas, on sites of pressure, following dermatomes or Blaschko lines, in intertriginous areas, in follicular localization

57

MORE DERM TERMS

- **Discoid** (nummular)
 - A filled circle, like a coin. (discoid eczema)
- **Arcuate**
 - Arcs, partial circles (urticaria)
- **Annular**
 - Circular, with different central skin compared with the rim (tinea corporis)
- **Serpiginous**
 - Snake-like (cutaneous larva migrans)
- **Livedo**
 - Altered (purple) skin color
- **Reticulate**
 - Fine, net-like pattern
 - Fifth disease = erythema infectiosum
 - Livedo reticularis
- **Target**
 - Concentric rings (erythema multiforme)
- **Linear**
 - Straight line (poison oak)

TARGET LESIONS



- Typically seen in erythema multiforme
- “Targetoid” lesions may appear in urticaria

SERPIGINOUS LESIONS



- Creeping, snake-like pattern
- Typically seen in:
 - Cutaneous larva migrans
 - Ringworm
 - Erythema annulare centrifugum

LIVEDO RETICULARIS



- A reticular, net-like pattern in the skin

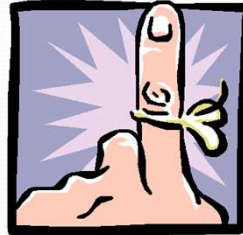
LICHENIFICATION



- Thickened, hyperpigmented skin with exaggerated skin lines

REMEMBER...

- Every time you describe a dermatologic finding, you need to include:
 1. Type of skin lesion (i.e., what is the primary and secondary morphology?)
 2. Color
 3. Margination
 4. Consistency
 5. Shape
 6. Arrangement
 7. Distribution of lesions
 8. Include a picture, whenever you can



CLINICAL SIGNS

Dermatographism

- The appearance of an urticarial wheal after focal pressure (e.g. stroking or scratching the skin) in the distribution of the pressure



Darier sign

- Refers to rapid swelling of a lesion when stroked
- May not appear for 5-10 minutes

Auspitz sign

- “Positive” when slight scratching or curetting of a scaly lesion reveals punctate bleeding points within the lesion
- Suggestive of psoriasis, but is not specific



CLINICAL SIGNS

Nikolsky phenomenon

- The epidermis is dislodged from the dermis by lateral, shearing pressure with a finger, resulting in an erosion
- Occurs in pemphigus, staphylococcal scalded skin (SSS) syndrome, and in epidermo-necrotic disorders, e.g. toxic epidermal necrolysis

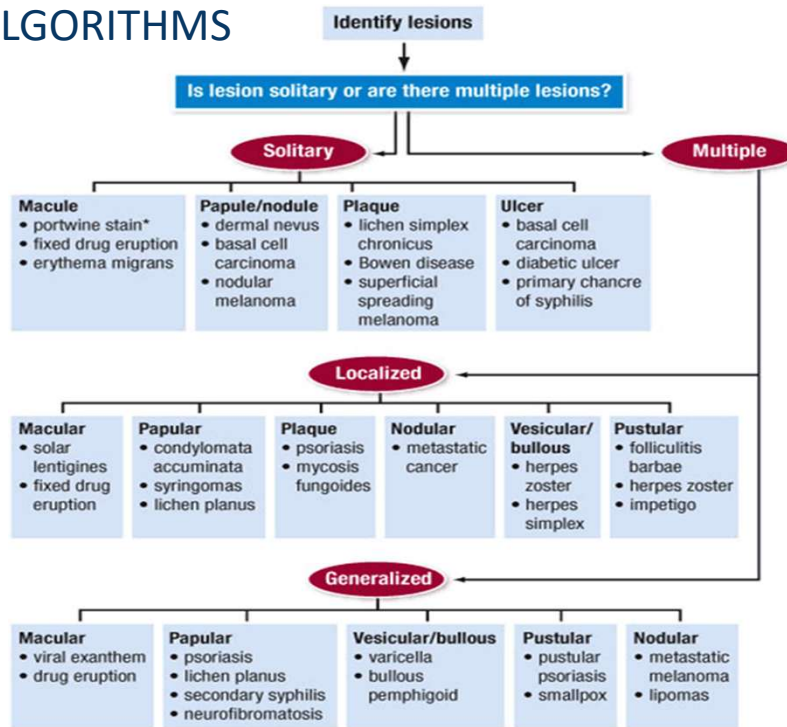


Koebner phenomenon

- The development of lesions within areas of trauma (e.g. caused by scratching, rubbing, injury)



USE ALGORITHMS



*Bulleted conditions are examples

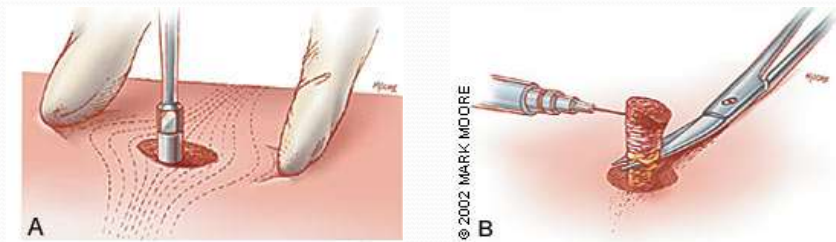
AT THE CONCLUSION OF YOUR METHODOICAL APPROACH

1. Formulate a differential diagnosis (DDX):

- This is a list of several possible diagnoses, with the most likely diagnosis at the top of the list
- This will help you decide what further testing needs to be done and what plan to initiate

2. Perform and order tests to confirm your chief diagnosis

- Biopsy, skin scraping, aspiration, microscopy, culture, etc.



3. Choose an appropriate treatment

- Topical and/or systemic therapy, surgical removal, etc.

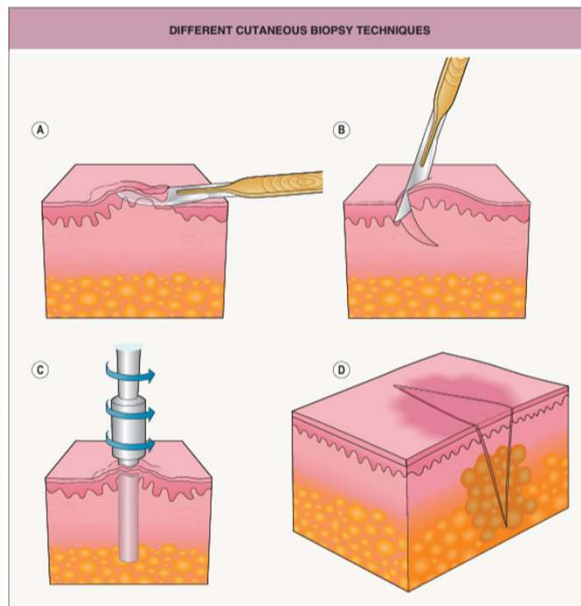
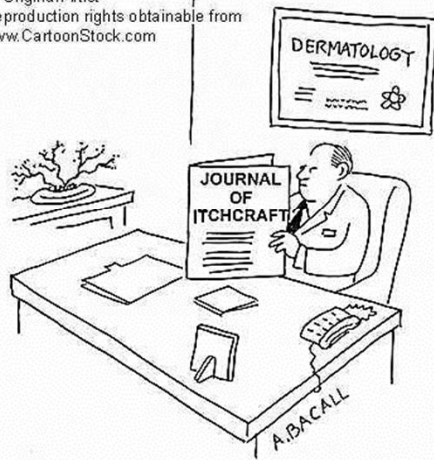


Fig. 0.13 Different cutaneous biopsy techniques. The size, topography, depth and site of the lesion, as well as the clinical differential diagnosis, influence the type of biopsy technique that is performed. **A** Superficial shave biopsy. **B** Deep shave biopsy (saucerization). **C** Punch biopsy. **D** Incisional biopsy. For more details, see text (also see [Ch. 146](#)). Courtesy Suzanne Olbricht, MD.

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